

Carewell Pharma - A Family of Learning

search CAREWELL PHARMA on YouTube & Watch Videos Syllabus wise

(1)

Topic **UNIT-IVth** - PHARMACOGNOSY-II Date _____

* Industrial production, Estimation & utilization of the following phytoconstituents.

1. forskolin:-

Biological sources:- It is obtained from dried plant roots of Coleus forskohlii belonging to Lamiaceae family

Industrial Production:-

→ Tuber of forskohlii are collected, washed & dried



Website :- www.carewellpharma.in

Visit Website & Download FREE Notes

Carewell Pharma - A Family of Learning

search CAREWELL PHARMA on YouTube & Watch Videos Syllabus wise

②

Topic _____

Date _____

↓
Dried roots powder is extracted using methanol as solvent in Soxhlet unit for 6 hours.

↓
filtered the methanolic extract & the resultant concentrated is added with chloroform.

↓
Equal volume of water is added to the separating funnel and treated 2-3 times.

↓
The chloroform layer is separated & concentrated

↓
A precipitate of forskolin is obtained by using ice-cold n-hexane.

↓
A Reddish brown or brown coloured powder is obtained.

Estimation:

By TLC & HPLC method.



Website :- www.carewellpharma.in

Visit Website & Download FREE Notes

Carewell Pharma - A Family of Learning

search CAREWELL PHARMA on YouTube & Watch Videos Syllabus wise

(3)

Topic_____

Date_____

Utilisation:-

- It is used in treatment of asthma, glaucoma & vasodilation.
- It also used in the treatment of insomnia, respiratory disorder & angina.

Website :- www.carewellpharma.in

Visit Website & Download FREE Notes

Carewell Pharma - A Family of Learning

search CAREWELL PHARMA on YouTube & Watch Videos Syllabus wise

④

Topic _____

Date _____

2. Sennosides:-

Biological sources:- Sennoside is obtained from the dried leaves of *Cassia angustifolia* belonging to legumin family.

Industrial Production:-

→ Senna leave are dried, powdered & extracted with benzene for 2 hrs. on electronic shaker to remove chlorophyll



Extract is filtered & marc is dried at room temp.



dried marc extracted with 70% methanol for 4-6 hrs.



Repeated twice i.e. extracted with methanol for 2 hrs & extract is filtered.



Then pH is adjusted to 3.2 by HCl & kept aside for 2 hrs, 5°C.



Later on CaCl_2 is added with regular stirring.



Carewell Pharma - A Family of Learning

search CAREWELL PHARMA on YouTube & Watch Videos Syllabus wise

③

Topic_____

Date_____

↓
Now pH is adjusted to 8 using ammonia set aside for 2 hr.

↓
Precipitate of sennoside are purified, filtered & dried over diphosphorous pentoxide in desiccator

↓
Percentage yield of sennoside is calculated.

Estimation:-

Column chromatography.

Stationary phase - C_{18}

Plate - Silica gel 60 f_{254}

Mobile phase - 1% Acetic acid in H_2O .

* Utilization:-

- Senna is used in treating constipation, also used as alternative medicine as a laxative.
- It is used in treating gout, anaemia, malaria, jaundice.

Carewell Pharma - A Family of Learning

search CAREWELL PHARMA on YouTube & Watch Videos Syllabus wise

⑥

Topic _____

Date _____

3 Artemisin:-

Biological source:- it is invented in china, a sesquiterpene obtained from leaves & flower heads of *Artemisia annua* belongs to family Asteraceae.

Industrial Production:-

→ *Artemisia annua* is grown & propagated in a hormone free medium by process of microcutting.

↓
It is extracted from air dried plant using ethylether, petroleum ether gasoline as a solvent.

↓
extraction with hexane at room temperature for several day is also effective.

↓
The fraction are conc. & dried, purified using column chromatography on silica gel absorbent

↓
Artemisin is obtained as fine white crystals.

Website :- www.carewellpharma.in

Visit Website & Download FREE Notes

Carewell Pharma - A Family of Learning

search CAREWELL PHARMA on YouTube & Watch Videos Syllabus wise

⑦

Topic _____

Date _____

Estimation:-

By HPTLC.

Mobile phase - n-hexane, ethylacetate

Stationary phase - Silicagel

Detector - UV 260 nm.

Utilization:-

- It is used as chinese medicine to treat fever, inflammation & malaria.
- It have ability to seppress inflammatory immune reactions.
- It also has anticancer activity, chronic bladder infection.

4. Diasgenin:-

Biological source:- It is obtained from tubers of Dioscorea deltoidea plants belongs to family Dioscoreaceae.

Industrial Production:- Can be extracted by any of two methas i.e acid hydrolysis & alce extraction method.

Website :- www.carewellpharma.in

Visit Website & Download FREE Notes

Carewell Pharma - A Family of Learning

search CAREWELL PHARMA on YouTube & Watch Videos Syllabus wise

(2)

Topic_____

Date_____

1 Acid hydrolysis method.

→ Dried ribosomes are powdered & subjected to hydrolysis by refluxing with 5% H_2O HCl for 2 hrs.



The hydrolyzed mass is filtered, washed twice with H_2O & then twice with 5% Na_2CO_3 soln.



finally, residue is obtained & extracted with toluene for 8 hrs.



extract is conc. during which disogenin gets ppt.



Disogenin is filtered, washed with little hexane & dried to yield about 95% pure products.

Estimation:

By TLC, HPTLC.

Stationary phase - Silica gel plates

Mobile phase - mixture of CHCl_3 , ethanol, CHCl_3 .



Carewell Pharma - A Family of Learning

search CAREWELL PHARMA on YouTube & Watch Videos Syllabus wise

⑨

Topic _____

Date _____

Utilization:-

- It is used to treating CHF, arrhythmia.
- It also used in treating precursor for steroidal synthesis
- It is used in preparation of sex hormones, oral contraceptives.

5. Digoxin:-

Biological source:- It is obtained from Digitalis Lanata belonging to Plant genaneae family.

Industrial Production:-

→ Powder drug is extracted with 70% ethanol and treated on hot plate for 20 min



Now, filtered the solution.



150 ml of water & acid solution is added to filtrate



Website :- www.carewellpharma.in

Visit Website & Download FREE Notes

Carewell Pharma - A Family of Learning

search CAREWELL PHARMA on YouTube & Watch Videos Syllabus wise

(10)

Topic_____

Date_____

↓
Precipitate of phenolic compound is obtained.

↓
10% of H_2SO_4 is added dropwise till no ppt form.

↓
the excess amount of lead ions in the form of insoluble lead sulphate is removed.

Estimation:-

By TLC

Stationary phase - Pre-coated silica gel

Detecting agent - perchloric acid.

Mobile phase - Ethanol

Sample - Plant extract.

Utilization:-

→ It is used in the treatment of arrhythmia.

→ It is used along with other medication for treating heart failure.



Website :- www.carewellpharma.in

Visit Website & Download FREE Notes

Carewell Pharma - A Family of Learning

search CAREWELL PHARMA on YouTube & Watch Videos Syllabus wise

(11)

Topic _____

Date _____

6. Atropine:-

Biological source:- It is obtained from the tops of plant Atropa belladonna belonging to Solanaceae family.

Industrial production:-

→ Powdered of leaves are extracted with 95% of ethanol



the ethanol is distilled off, & treating the obtained syrupy mass with 1% HCl.



acidified solution is treated with light petroleum ether.



add ammonia to make it alkaline & treated with chloroform.



The chloroform layer is separated & evaporated



Website :- www.carewellpharma.in

Visit Website & Download FREE Notes

Carewell Pharma - A Family of Learning

search CAREWELL PHARMA on YouTube & Watch Videos Syllabus wise

12

Topic _____

Date _____

↓
Oxalic acid is added to get oxalates of atropine & hyoscyamine.

↓
Separated the atropine from hyoscyamine by fractional crystallisation.

Estimation:-



Website :- www.carewellpharma.in

Visit Website & Download FREE Notes

Carewell Pharma - A Family of Learning

search CAREWELL PHARMA on YouTube & Watch Videos Syllabus wise

Topic _____

Date _____

7. Podophyllotoxin:-

Biological sources:- It is a resin obtained from dried roots & rhizomes of plant *podophyllum hexan drum*

Podophyllum peltatum & *Podophyllum emodii* belonging to berberidaceae.

Industrial Production:-

→ The dried root of & rhizomes of *Podophyllum* is extracted with methanol



methanol extract is filtered & evaporated to semi-solid mass, which is then dissolved in acidic H_2O result into ppt. of Podophyllotoxin.



The ppt. allowed to settle for 2 hrs & filtered & washed with cold water to remove acidic fraction, dried to get dark powder.



further, purified by dissolving in alcohol to get *podophyllum* & recrystallized using benzene to yield pure *podophyllum*.



Website :- www.carewellpharma.in

Visit Website & Download FREE Notes

Carewell Pharma - A Family of Learning

search CAREWELL PHARMA on YouTube & Watch Videos Syllabus wise

(14)

Topic_____

Date_____

Estimation:-

By HPLC

Mobile phase :- Methanol, water.

Detection :- 280 nm.

Column :- Cis-Symmetry.

Utilization:-

- It have anti-cancer activity like its derivative eloposide used in ovarian, lung, testes, lymphoma cancer.
- It is used as purgative, to treat constipation.
- It also have antitumor activity.

8. Caffeine.

Biological source:-



Website :- www.carewellpharma.in

Visit Website & Download FREE Notes

Carewell Pharma - A Family of Learning

search CAREWELL PHARMA on YouTube & Watch Videos Syllabus wise

(15)

Topic _____

Date _____

9. Taxol :-

Biological source :- It is obtained from bark of plants is Taxus baccifolia, Taxus canadensis, belonging to family Taxaceae.

Industrial Production :-

→ The dried ground bark is extracted with methanol

↓
obtained concentrate is again extracted with dichloromethane & the solvent extract is conc. to a powder

↓
The powder is stirred with mixture of acetone & filtered to remove the insoluble water.

↓
Partition with mixture of CCl_4 & evaporated

↓
dried CCl_4 fraction again extracted with CCl_4 , methanol to obtain Crude taxol.

Website :- www.carewellpharma.in

Visit Website & Download FREE Notes

Carewell Pharma - A Family of Learning

search CAREWELL PHARMA on YouTube & Watch Videos Syllabus wise

(16)

Topic _____

Date _____

Estimation:-

By HPTLC method.

- Mobile phase - Chloroform, methanol (7:1)
- Visualizing agent - Vanillin sulphuric acid.

Utilization:-

- It is used in treatment of ovarian, lung cancer.
- It is also used as antiproliferative agent.
- It is used in treatment of bladder & esophageal cancer.



Website :- www.carewellpharma.in

Visit Website & Download FREE Notes